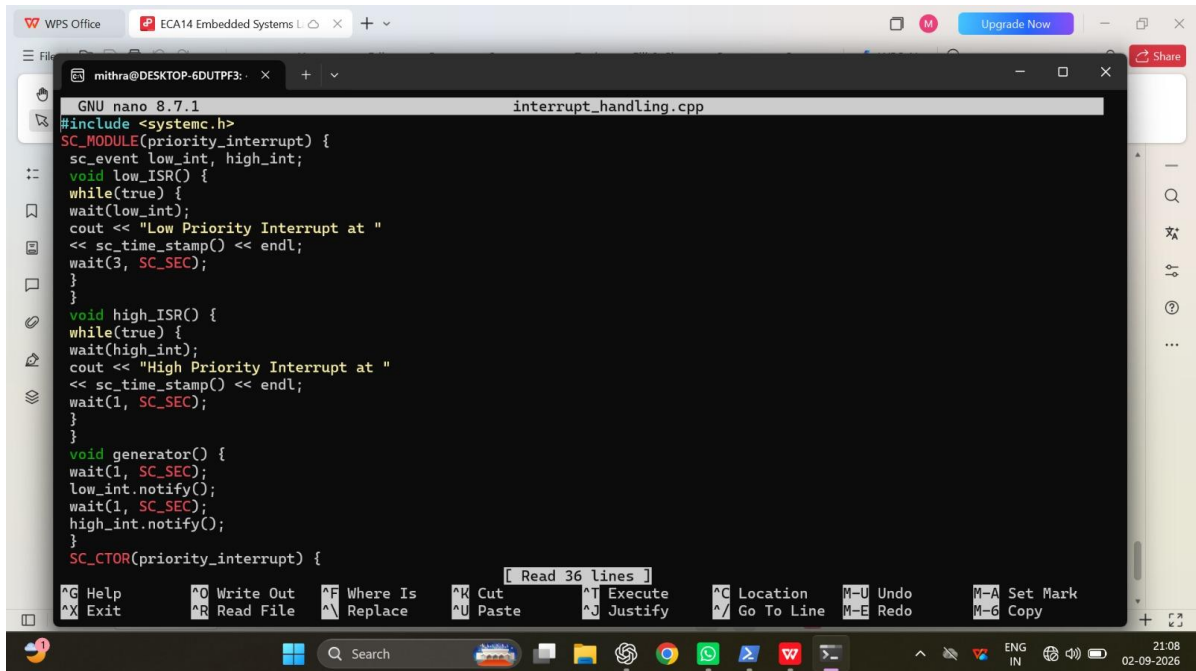


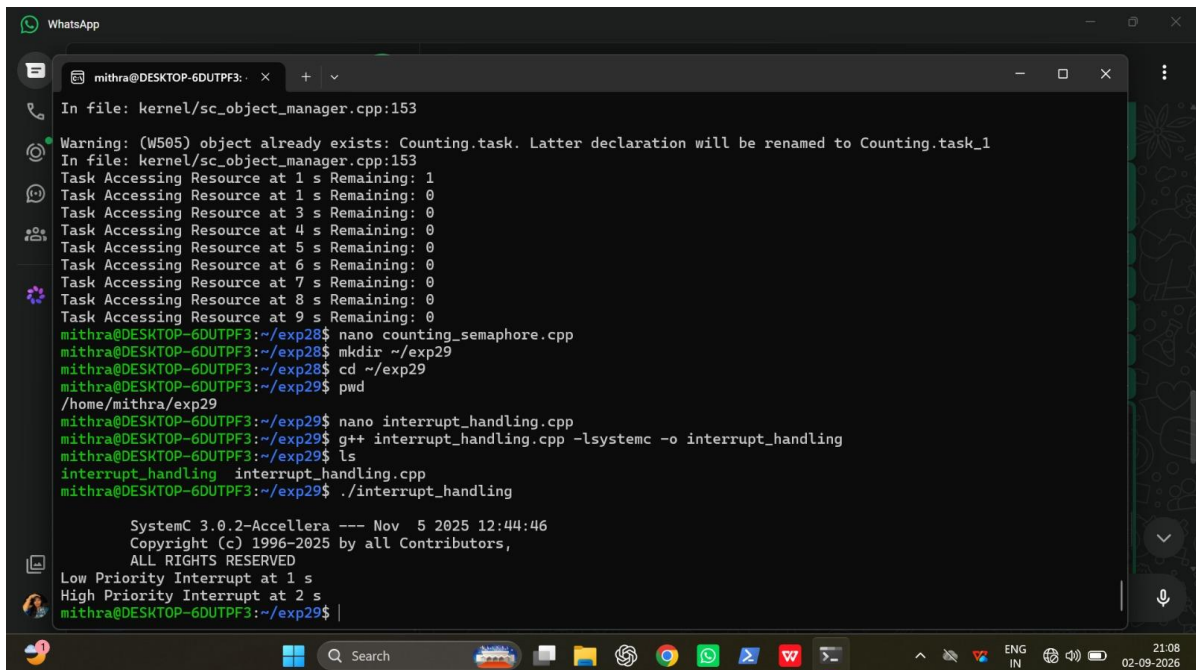
Exp No: 29

PRIORITY-BASED INTERRUPT HANDLING



```
GNU nano 8.7.1 interrupt_handling.cpp
#include <systemc.h>
SC_MODULE(priority_interrupt) {
    sc_event low_int, high_int;
    void low_ISR() {
        while(true) {
            wait(low_int);
            cout << "Low Priority Interrupt at "
            << sc_time_stamp() << endl;
            wait(3, SC_SEC);
        }
    }
    void high_ISR() {
        while(true) {
            wait(high_int);
            cout << "High Priority Interrupt at "
            << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }
    void generator() {
        wait(1, SC_SEC);
        low_int.notify();
        wait(1, SC_SEC);
        high_int.notify();
    }
    SC_CTOR(priority_interrupt) {

```



```
WhatsApp
mithra@DESKTOP-6DUTPF3: ~
In file: kernel/sc_object_manager.cpp:153
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1
In file: kernel/sc_object_manager.cpp:153
Task Accessing Resource at 1 s Remaining: 1
Task Accessing Resource at 1 s Remaining: 0
Task Accessing Resource at 3 s Remaining: 0
Task Accessing Resource at 4 s Remaining: 0
Task Accessing Resource at 5 s Remaining: 0
Task Accessing Resource at 6 s Remaining: 0
Task Accessing Resource at 7 s Remaining: 0
Task Accessing Resource at 8 s Remaining: 0
Task Accessing Resource at 9 s Remaining: 0
mithra@DESKTOP-6DUTPF3:~/exp28$ nano counting_semaphore.cpp
mithra@DESKTOP-6DUTPF3:~/exp28$ mkdir ~/exp29
mithra@DESKTOP-6DUTPF3:~/exp28$ cd ~/exp29
mithra@DESKTOP-6DUTPF3:~/exp29$ pwd
/home/mithra/exp29
mithra@DESKTOP-6DUTPF3:~/exp29$ nano interrupt_handling.cpp
mithra@DESKTOP-6DUTPF3:~/exp29$ g++ interrupt_handling.cpp -lsystemc -o interrupt_handling
mithra@DESKTOP-6DUTPF3:~/exp29$ ls
interrupt_handling  interrupt_handling.cpp
mithra@DESKTOP-6DUTPF3:~/exp29$ ./interrupt_handling

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
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ALL RIGHTS RESERVED
Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s
mithra@DESKTOP-6DUTPF3:~/exp29$
```